Type MS · Log equations that turn into a quadratic

Answer 1 of 8 · Question 8 part whole · 7 marks

2020 NOV · Paper 02 (regional R) · extract 1 of 2 · find part whole in this mark scheme extract

8	$\log_x 3 = \dfrac{1}{1\log_3 x}$	B1
	Let $y = \log_3 x$	
	So $y - \dfrac{2}{y} = 1$	M1
	$y^2 - y - 2 = 0$	A1
	$(y-2)(y+1) = 0$	M1
	$\log_3 x = 2$ or $\log_3 x = -1$	M1
	$x = 9$ or $x = \dfrac{1}{3}$	A1 A1
		[7]

	Notes
B1	For use of $\log_a x = \dfrac{1}{\log_b a}$
M1	For $y - \dfrac{2}{y} = 1$ oe
A1	For rearranging to a 3 TQ
M1	For solving the 3 TQ
M1	For either $\log_3 x = 2$ or $\log_3 x = -1$
A1	For $x = 9$
A1	For $x = \dfrac{1}{3}$

Type MS · Log equations that turn into a quadratic

Answer 1 of 8 · Question 8 part whole · 7 marks

2020 NOV · Paper 02 (regional R) · extract 2 of 2 · find part whole in this mark scheme extract

Question number	Scheme	Marks

Type MS · Log equations that turn into a quadratic

Answer 2 of 8 · Question 5 part b · 7 marks

2020 JAN · Paper 01 (regional R) · extract 1 of 2 · find part b in this mark scheme extract

	[illegible]
M1	$f'(2) = 5$ leading to an equation in a and b
A1	$4a + 2b + 34 = 0$ and $4a + b + 19 = 0$
M1	Solving simultaneously
A1	$b = -15,\ a = -1$
(b)	
M1	Dividing by $x - 2$ to obtain a 3TQ
M1	Factorising the 3TQ
A1	All 3 terms correct
(c)	
B2 ft	$x = 2,\ -3,\ \frac{3}{2}$ (**B1** for 2 correct)

Type MS · Log equations that turn into a quadratic

Answer 2 of 8 · Question 5 part b · 7 marks

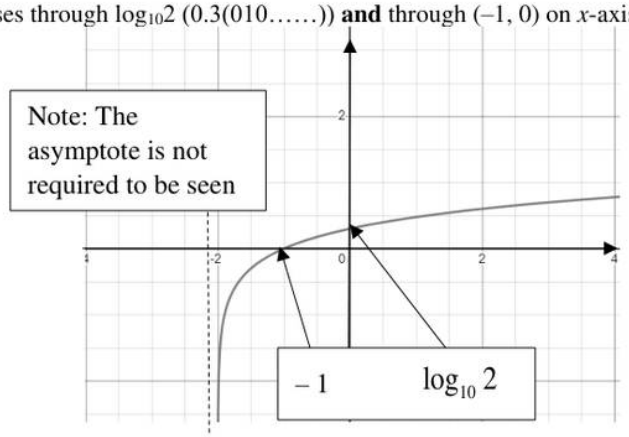
2020 JAN · Paper 01 (regional R) · extract 2 of 2 · find part b in this mark scheme extract

Question number	Scheme	Marks
5 (a)	$\log_4 32 = \dfrac{\log_2 32}{\log_2 4} = \dfrac{5}{2}$* or $\log_4 32 = \log_4 4^{\frac{5}{2}} = \dfrac{5}{2}$* or $\log_4 32 = \log_{2^2} 2^5 = \dfrac{5}{2}$* **ALT** $\log_4 32 = a \quad \Rightarrow 4^a = 32 \quad \Rightarrow a = \dfrac{5}{2}$ *	M1A1cso [2] {M1}{A1} cso [2]
(b)	$\log_2 x - \log_4 32 + \dfrac{1}{4}\log_x 16 = 0$ Let $\log_2 x = y$ $y - \dfrac{5}{2} + \dfrac{1}{4}\left(\dfrac{\log_2 16}{\log_2 x}\right) = 0$ or $y - \dfrac{5}{2} + \dfrac{1}{\log_2 x} = 0$ $\Rightarrow y - \dfrac{5}{2} + \dfrac{1}{y} = 0$ $\Rightarrow 2y^2 - 5y + 2 = 0$ $\Rightarrow (2y-1)(y-2) = 0$ $\Rightarrow y = \log_2 x = \dfrac{1}{2}$ or 2 $\Rightarrow x = 2^{\frac{1}{2}} = \sqrt{2}$ and $x = 2^2 = 4$	M1 M1A1 M1 M1 M1A1 [7]
		Total 9 marks

Type MS · Log equations that turn into a quadratic

Answer 3 of 8 · Question 7 part c · 6 marks

2022 JUN · Paper 01 (regional R) · extract 1 of 3 · find part c in this mark scheme extract

7 (a)	Shape of the curve, **crossing** negative x-axis and positive y-axis with the asymptotic nature of the curve shown. passes through $\log_{10}2$ (0.3(010……)) **and** through (–1, 0) on x-axis  	B1 B1 [2]
(b)	$\left(2\log_a 4+2\log_a 4^2\right)$ or $(2\log_a 2^2+2\log_a 2^4)$ $2\log_a 4+4\log_a 4$ or $4\log_a 2+8\log_a 2$ $\log_a 4=\frac{1}{6}$ or $\log_a 2=\frac{1}{12}$ oe $a^{\frac{1}{6}}=4$ or $a^{\frac{1}{12}}=2$ $a=4096$ **ALT** $(2)\log_a 64=1\Rightarrow\log_a 64=\frac{1}{2}$ $\Rightarrow a^{\frac{1}{2}}=64$ $\Rightarrow a=4096$	M1 dM1 A1 [M1 dM1 A1] [3]
(c)	$\log_q 16\rightarrow\frac{\log_2 16}{\log_2 q}$ or $\log_2 q\rightarrow\frac{\log_q q}{\log_q 2}$ or $\frac{1}{\log_q 2}$ $\left(5\frac{\log_2 16}{\log_2 q}+4\log_2 q=24\right)$ or $\left(5\log_q 16+4\frac{\log_q q}{\log_q 2}=24\right)$ $20+4(\log_2 q)^2=24\log_2 q$ or $20(\log_q 2)^2+4=24\log_q 2$ $(\log_2 q-5)(\log_2 q-1)(=0)$ or $(5\log_q 2-1)(\log_q 2-1)(=0)$ $\left(\log_2 q=5\right)\rightarrow q=32$ or $\left(\log_q 2=\frac{1}{5}\right)\rightarrow q=32$ $\left(\log_2 q=1\right)\rightarrow q=2$ or $\left(\log_q 2=1\right)\rightarrow q=2$	M1 M1 A1 dM1 M1 A1 [6]
		Total 11 marks

Type MS · Log equations that turn into a quadratic

Answer 3 of 8 · Question 7 part c · 6 marks

2022 JUN · Paper 01 (regional R) · extract 2 of 3 · find part c in this mark scheme extract

Part	Mark	Additional Guidance
(a)	B1	The curve must not bend back on itself and should cross both axes once on negative x and once on positive y. It needs to demonstrate an asymptotic nature. Accept an asymptote drawn correctly as well as their curve.
	B1	It is enough for -1 to be indicated on the x-axis with the curve passing through this point and for $\log_{10}2$ to be marked on the y- axis (decimal 0.3…… allowed) There must be a curve drawn (even incorrectly) to achieve this mark. **Do not** accept intersections unless they are marked on the graph in the correct place.
(b)	M1	For the use of the power rule with logs to obtain $2\log_a 4$ from $\log_a 16$ OR $4\log_a 2$ from $\log_a 16$
	dM1	A **correct** rearrangement of their equation to obtain for example $a^{\frac{1}{6}} = 4$ or $a^{\frac{1}{12}} = 2$ or even of the type $a^1 = 64^2$ There are many ways of completing this. Look for a correct value for the corresponding correct power of a. **This mark is dependent on the first M mark**. Look for correct log work for their values.
	A1	For $a = 4096$
ALT	M1	For use of the correct log rule to obtain $\log_a 64$
	dM1	A correct rearrangement of their equation to obtain $a^{\frac{1}{2}} = 64$ **This mark is dependent on the first M mark**. Look for correct log work for their values.
	A1	For $a = 4096$
(c)	M1	For a fully correct change of base of log to either base 2 or base q
	M1	For an attempt to rearrange and form a 3TQ quadratic. Allow one arithmetical slip only.
	A1	For the correct 3TQ.
	dM1	**Dependent on the second M mark** – an acceptable attempt to solve the quadratic equation – see general guidance.
	M1	For a correct application using either of their solutions moving from log form to exponential form. **Note this is an independent mark.**
	A1	**Both** correct solutions.

Type MS · Log equations that turn into a quadratic

Answer 3 of 8 · Question 7 part c · 6 marks

2022 JUN · Paper 01 (regional R) · extract 3 of 3 · find part c in this mark scheme extract

Question number	Scheme	Marks

Type MS · Log equations that turn into a quadratic

Answer 4 of 8 · Question 6 part whole · 8 marks

2023 JUN · Paper 01 · extract 1 of 3 · find part whole in this mark scheme extract

6	$\log_2 x^3 + \log_4 x^2 - 3\log_x 2 = \log_2 x^3 + \dfrac{\log_2 x^2}{\log_2 4} - \dfrac{3\log_2 2}{\log_2 x} = 0$	M1
	$= 3\log_2 x + \dfrac{2\log_2 x}{\log_2 4} - \dfrac{\log_2 2^3}{\log_2 x} = 0$	M1
	$3\log_2 x + \dfrac{2\log_2 x}{2} - \dfrac{3}{\log_2 x} = 0$	B1
	$\Rightarrow 3(\log_2 x)^2 + (\log_2 x^2)^2 - 3 = 0$	M1
	$\Rightarrow 4(\log_2 x)^2 = 3 \Rightarrow (\log_2 x)^2 = \dfrac{3}{4} \Rightarrow \log_2 x = \pm\sqrt{\dfrac{3}{4}}$	M1
	$\Rightarrow x = 2^{\sqrt{\frac{3}{4}}} \approx 1.82$ or $x = 2^{-\sqrt{\frac{3}{4}}} \approx 0.549$	M1A1A1
		[8]
		Total 8 marks

Mark	Notes NB: Candidates will frequently use a substitution for their chosen log.
Working in log base 2 **Correct answer/s with no working scores no marks.**	
M1	For changing the base of the log correctly in at least one term $\log_2 x^3 + \log_4 x^2 - 3\log_x 2 = \log_2 x^3 + \dfrac{\log_2 x^2}{\log_2 4} - \dfrac{3\log_2 2}{\log_2 x} = 0$
M1	For using the power law in at least one term. $\log_2 x^3 = 3\log_2 x$ or $\log_2 x^2 = 2\log_2 x$ or $3\log_2 2 = \log_2 2^3$ $\Rightarrow 3\log_2 x + \log_2 x - \dfrac{\log_2 2^3}{\log_2 x} = 0$
B1	For either $\log_2 4 = 2$ or $\log_2 8 = 3$
M1	For multiplying through by $\log_2 x$ $3(\log_2 x)^2 + (\log_2 x)^2 - 3 = 0$
M1	For simplifying and obtaining two values for $\log_2 x$ using a valid method. **NB:** If they discard one value at any stage do not award this mark. $4(\log_2 x)^2 - 3 = 0 \Rightarrow \log_2 x = \pm\sqrt{\dfrac{3}{4}}$
M1	For removing the log to find at least one value for x $x = 2^{\sqrt{\frac{3}{4}}}$ or $x = 2^{-\sqrt{\frac{3}{4}}}$
A1	For either awrt $x = 1.82$ or 0.549
A1	For awrt both $x = 1.82$ and 0.549

Type MS · Log equations that turn into a quadratic

Answer 4 of 8 · Question 6 part whole · 8 marks

2023 JUN · Paper 01 · extract 2 of 3 · find part whole in this mark scheme extract

Working in log base 4	
M1	For changing the base of the log correctly in at least one term $\log_2 x^3 + \log_4 x^2 - 3\log_x 2 = \frac{\log_4 x^3}{\log_4 2} + \log_4 x^2 - \frac{3\log_4 2}{\log_4 2} = 0$
M1	For using the power law in at least one term $\frac{3\log_4 x}{\log_4 2} + 2\log_4 x - \frac{3\log_4 2}{\log_4 2} = 0$
B1	For either $\log_4 2 = \frac{1}{2}$ or $\log_4 8 = \frac{3}{2}$
M1	For multiplying through by $\log_4 x$ $\qquad 6(\log_4 x)^2 + 2(\log_4 x^2)^2 - \frac{3}{2} = 0$
M1	For simplifying and obtaining two values for $\log_4 x$ $8(\log_4 x)^2 = \frac{3}{2} \Rightarrow (\log_4 x)^2 = \frac{3}{16} \Rightarrow \log_4 x = \pm\sqrt{\frac{3}{16}}$ **NB:** If they discard one value at any stage do not award this mark.
M1	For removing the log to find at least one value for x using a valid method. $x = 4^{\sqrt{\frac{3}{16}}} \approx \ldots$ or $x = 4^{-\sqrt{\frac{3}{16}}} \approx \ldots$
A1	For either awrt $x = 1.82$ or 0.549
A1	For awrt both $x = 1.82$ and 0.549
Working in log base x	
M1	For changing the base of the log correctly in at least one term $\log_2 x^3 + \log_4 x^2 - 3\log_x 2 = \frac{\log_x x^3}{\log_x 2} + \frac{\log_x x^2}{\log_x 4} - 3\log_x 2 = 0$
M1	For using the power law in at least one term $\frac{3\log_x x}{\log_x 2} + \frac{2\log_x x}{2\log_x 2} - 3\log_x 2 = 0$
B1	For $\log_x x = 1$ $\qquad \frac{3}{\log_x 2} + \frac{1}{\log_x 2} - 3\log_x 2 = 0 \Rightarrow \frac{4}{\log_x 2} - 3\log_x 2 = 0$
M1	For multiplying through by $\log_x 2$ $4 - 3(\log_x 2)^2 = 0$
M1	For obtaining two values of $\log_x 2$ using a valid method $\qquad \left(\log_x 2 = \pm\sqrt{\frac{4}{3}}\right)$ **NB:** If they discard one value at any stage do not award this mark.
M1	For removing the log to find at least one value for x. $2 = x^{\sqrt{\frac{4}{3}}} \Rightarrow x = 2^{\sqrt{\frac{3}{4}}} \approx \ldots$ or $2 = x^{-\sqrt{\frac{4}{3}}} \Rightarrow x = 2^{-\sqrt{\frac{3}{4}}} \approx \ldots$
A1	For either awrt $x = 1.82$ or 0.549

Type MS · Log equations that turn into a quadratic

Answer 4 of 8 · Question 6 part whole · 8 marks

2023 JUN · Paper 01 · extract 3 of 3 · find part whole in this mark scheme extract

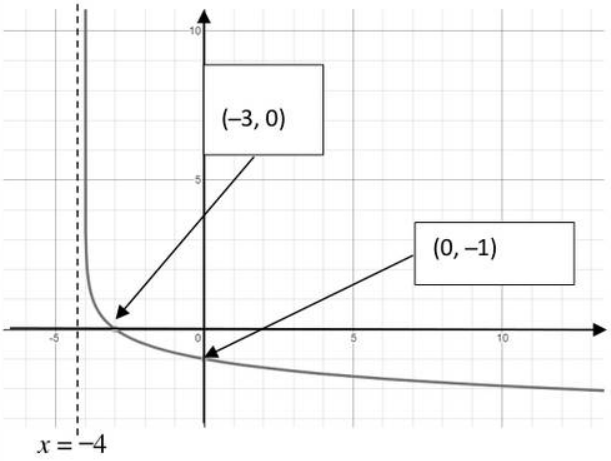
A1	For awrt both $x = 1.82$ and 0.549

Question	Scheme	Marks

Type MS · Log equations that turn into a quadratic

Answer 5 of 8 · Question 7 part b · 5 marks

2024 JUN · Paper 01 (regional R) · find part b in this mark scheme extract

7(a)		B1 Shape and position B1 Intersection with x-axis B1 Intersection with y-axis B1 Asymptote [4]
(b)	$\log_{(x+4)} 256 - \log_4(x+4) = 0 \Rightarrow \dfrac{\log_4 256}{\log_4(x+4)} - \log_4(x+4) = 0$ $\Rightarrow 4 - (\log_4(x+4))^2 = 0 \Rightarrow (\log_4(x+4))^2 = 4$ $\Rightarrow \log_4(x+4) = \pm 2$ $x+4 = 4^2 \Rightarrow x = 12$ $x+4 = 4^{-2} \Rightarrow x = -\dfrac{63}{16}$ or -3.9375	M1 M1 M1 A1 A1 [5]
		Total 9 marks

Type MS · Log equations that turn into a quadratic

Answer 6 of 8 · Question 9 part ii · 6 marks

2025 JUN · Paper 02 (regional R) · extract 1 of 3 · find part ii in this mark scheme extract

Number		
9 (i)	$\left(3(\log_a 9+\log_a 27)=1\right)$	
	$3(2\log_a 3+3\log_a 3)=1$ oe eg $2\log_a 3+3\log_a 3=\frac{1}{3}$	M1
	$\log_a 3=\frac{1}{15}$	
	$a^{\frac{1}{15}}=3$ or $\log_3 a=15$	M1
	$(a=)3^{15}$	A1
	ALT	[3]
	$3(\log_a(9\times 27)=1$ oe eg $3(\log_a(3^5))=1$	
	$\log_a(9\times 27)=\frac{1}{3}$ oe eg $\log_a(3^5)=\frac{1}{3}$	M1
	$a^{\frac{1}{3}}=9\times 27$	M1
	$(a=)3^{15}$	A1
		[3]
(ii)	$\left(\log_4 p+\log_p 256=-4\right)$	
	$\log_4 p+\frac{\log_4 256}{\log_4 p}=-4$ or $\frac{\log_p p}{\log_p 4}+\log_p 256=-4$	M1
	$(\log_4 p)^2+\log_4 256=-4\log_4 p$ or $(\log_p p)+(\log_p 256)(\log_p 4)=-4\log_p 4$	dM1
	$(\log_4 p)^2+4\log_4 p+4(=0)$ or $4(\log_p 4)^2+4\log_p 4+1(=0)$	A1
	$((\log_4 p)+2)^2(=0)$ or $((2\log_p 4)+1)^2(=0)$	M1
	$\log_4 p=-2\rightarrow p=$ or $\log_p 4=-\frac{1}{2}\rightarrow p=$	M1
	$(p=)\frac{1}{16}$ oe	A1
		[6]
		Total 9 marks

Type MS · Log equations that turn into a quadratic

Answer 6 of 8 · Question 9 part ii · 6 marks

2025 JUN · Paper 02 (regional R) · extract 2 of 3 · find part ii in this mark scheme extract

Part	Mark	Additional Guidance
When marking (i) stick to 1 ALTERNATIVE only – eg it's not possible to gain M2 for an application of a power rule and an addition rule with logs. **Look out for equivalent working, particularly the use of** $\log_{a^b} c = \frac{1}{b}\log_a c$ **Send any work that you may think uses a different method worth partial marks to Review**		
(i)	M1	For correct use of the rule $\log_a A^b = b\log_a A$ This mark may be awarded anywhere the power rule is correctly used **once** in working, in either direction, even if there's other contradictory work.
	M1	For a rearrangement of $3(b\log_a 3 + c\log_a 3) = 1$ to $a^{\frac{1}{3R}} = 3$ or $\log_3 a = 3R \quad R > 1$ seen anywhere in the working, even if there's other contradictory work. Allow errors in rearrangement. May be implied by any correct value for a not in the required form eg 14348907, $\left(3^5\right)^3$, $\left(3^3\right)^5$, $9^{7.5}$
	A1	$(a =) 3^{15}$
ALT	M1	For correct use the rule $\log_a A + \log_a B = \log_a AB$ This mark may be awarded anywhere the addition rule is correctly used **once** in working, in either direction, even if there's other contradictory work.
	M1	For a rearrangement of $3(\log_a (C \times D)) = 1$ *to* $a^{\frac{1}{3}} = CD$ seen anywhere in the working, even if there's other contradictory work. Allow errors in rearrangement. May be implied by any correct value for x not in the required form eg 14348907, $\left(3^5\right)^3$, $\left(3^3\right)^5$, $9^{7.5}$
	A1	$(a =) 3^{15}$
(ii)	M1	Correct use of change of base of logs. This mark may be awarded anywhere a change of base of logs is used correctly **once** in working, even if there's other contradictory work.
	dM1	Rearranges their equation to a 3 term quadratic (= 0). Allow errors in rearrangement so long as reach an expression of the form $p(\log_4 p)^2 + q\log_4 p + r = 0$ or $s(\log_p 4)^2 + t\log_p 4 + u = 0 \quad p,q,r,s,t,u \neq 0$
	A1	Correct 3 term quadratic(= 0) as shown.
	M1	A minimally acceptable attempt to solve their 3 term quadratic, see general guidance. Students may have used an alternative variable here for $\log_4 p$ or $\log_p 4$ The mark is for a minimally acceptable attempt to solve a 3TQ – see general guidance. The correct value from the correct quadratic implies this mark, otherwise method must be shown.
	M1	Correctly coverts their log equation into a value for p. ie $\log_4 p = F \rightarrow p =$ or $\log_p 4 = G \rightarrow p =$
	A1	For $(p =) \frac{1}{16}$ oe

Type MS · Log equations that turn into a quadratic

Answer 6 of 8 · Question 9 part ii · 6 marks

2025 JUN · Paper 02 (regional R) · extract 3 of 3 · find part ii in this mark scheme extract

Question number	Scheme	Marks

Type MS · Log equations that turn into a quadratic

Answer 7 of 8 · Question 6 part ii · 7 marks

2024 NOV · Paper 02 · extract 1 of 3 · find part ii in this mark scheme extract

6 (i)	$5(2\log_b 3+\log_b 3)=3$	M1
	$\left(3\log_b 3=\frac{3}{5}\right)\Rightarrow \log_b 3=\frac{1}{5}$	M1
	$b^{\frac{1}{5}}=3$	M1
	$b=243$ or 3^5	A1
	ALT	
	$5\log_b 27=3$	M1
	$\log_b 27=\frac{3}{5}$	M1
	$b^{\frac{3}{5}}=27$	M1
	$b=243$ or $b=27^{\frac{5}{3}}$	A1 [4]
(ii)	$3\log_3 x+3\frac{\log_3 27}{\log_3 x}=,28$	M1,B1
	$3(\log_3 x)^2+9=28\log_3 x$	dM1
	$3(\log_3 x)^2-28\log_3 x+9(=0)\Rightarrow(3\log_3 x-1)(\log_3 x-9)(=0)$	ddM1
	$(\log_3 x=)\frac{1}{3}$ and $(\log_3 x=)9$	A1
	$x=3^{\frac{1}{3}},3^9$ oe exact form	M1 A1 [7]
		Total 11 marks

Type MS · Log equations that turn into a quadratic

Answer 7 of 8 · Question 6 part ii · 7 marks

2024 NOV · Paper 02 · extract 2 of 3 · find part ii in this mark scheme extract

Part	Mark	Additional Guidance
(i)	M1	Rewrites 9 as 3^2 and correctly uses the power log rule as shown.
	M1	Collects terms as rearranges to give $\log_b p = r$, allow 1 error in rearrangement.
	M1	Correctly converts their log equation to an exponential.
	A1	Cao
ALT	M1	Uses the log addition law correctly to combine into a single log.
	M1	Collects terms as rearranges to give $\log_b p = r$, allow 1 error in rearrangement.
	M1	Rearranges to the form $b^{\frac{p}{q}} = r$, allow 1 error in rearrangement.
	A1	Cao
(ii)	B1	Replacement of $8\log_4 128 = 28$, wherever we see
	M1	For a correct change of the same base of logs on the lhs of the equation e.g. $3\log_3 x + 3\dfrac{\log_3 27}{\log_3 x} = 28$ or $\dfrac{3}{\log_x 3} + 3\log_x 27 = 28$ Allow slips on the coefficients
	dM1	For forming a 3-term quadratic in any form e.g. $3(\log_3 x)^2 + 9 = 28\log_3 x$ or $3 + 9(\log_x 3)^2 = 28(\log_x 3)$
	ddM1	For solving their 3-term quadratic using a valid method – see general guidance. (Correct answers from a correct 3TQ scores this mark)
	A1	For $(\log_3 x =)\dfrac{1}{3}$ and $(\log_3 x =)9$ or $(\log_x 3) = 3$ and $(\log_x 3) = \dfrac{1}{9}$
	M1	For correctly rearranging either of their equations, removes logs to find $x =$
	A1	For both correct values oe exact form. E.g. $x = 19683, x = \sqrt[3]{3}$ **isw**

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2024 NOV · Paper 02 · extract 3 of 3 · find part ii in this mark scheme extract

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Answer 8 of 8 · Question 10 part whole · 7 marks

2023 JUN · Paper 02 (regional R) · extract 1 of 3 · find part whole in this mark scheme extract

10	$8\log_x 64 = \dfrac{8\log_4 64}{\log_4 x}$	M1
	$\log_4 x^3 = 3\log_4 x$	M1
	$\log_4 x^3 + 8\log_x 64 = 22 \Rightarrow 3\log_4 x + \dfrac{8\log_4 64}{\log_4 x} = 22$	
	$\Rightarrow 3(\log_4 x)^2 + 8\log_4 64 = 22\log_4 x \Rightarrow 3(\log_4 x)^2 - 22\log_4 x + 24 = 0$	M1
	$3(\log_4 x)^2 - 22\log_4 x + 24 = 0 \Rightarrow (3\log_4 x - 4)(\log_4 x - 6) = 0$	M1
	$\Rightarrow \log_4 x = \dfrac{4}{3},\quad 6$	A1
	$x = 4^{\frac{4}{3}}$ or awrt 6.35 and $x = 4096$	M1A1 [7]
		Total 7 marks

NOTE WELL! This can be solved using a modern calculator. No working = no marks. Award marks only for work explicitly seen.

Mark	Notes
Works in base 4	
M1	For changing the base of the log correctly to base 4 $3\log_4 x + 8\log_x 64 = 22 \Rightarrow 3\log_4 x + \dfrac{8\log_4 4^3}{\log_4 x} = 22$
M1	For applying the power law correctly seen anywhere in their work. This mark can also be awarded for explicit application of the power law on $\log_4 64 = 3\log_4 4$
M1	For multiplying through by $\log_4 x$ and forming a 3TQ in log base 4
M1	For solving their 3TQ by any valid and correct method. If there is no method seen with an incorrect 3TQ or with incorrect solutions following a correct 3TQ this is M0 They must obtain two values for their log.
A1	For both correct values of $\log x$ $\left[\dfrac{4}{3},\quad 6\right]$
M1	For undoing either log correctly. Allow this mark for any erroneous log they find, but undo correctly.
A1	For **both** values of x Accept $4^{\frac{4}{3}}$ or awrt 6.35 and 4^6 or 4096

Type MS · Log equations that turn into a quadratic

Answer 8 of 8 · Question 10 part whole · 7 marks

2023 JUN · Paper 02 (regional R) · extract 2 of 3 · find part whole in this mark scheme extract

Works in base x	
M1	For changing the base of the log correctly to base x or vice versa. $\log_4 x^3 + 8\log_x 64 = 22 \Rightarrow \dfrac{3\log_x x}{\log_x 4} + 8\log_x 4^3 = 22$
M1	For applying the power law correctly seen anywhere in their work. This mark can also be awarded for explicit application of the power law on $\log_x 64 = 3\log_x 4$
M1	For multiplying through by $\log_x 4$ and forming a 3TQ in log base x $\dfrac{3}{\log_x 4} + 24\log_x 4 = 22 \Rightarrow 24(\log_x 4)^2 - 22\log_x 4 + 3 = 0$
M1	For solving their 3TQ by any valid and correct method. If there is no method seen with an incorrect 3TQ or with incorrect solutions following a correct 3TQ this is M0 $24(\log_x 4)^2 - 22\log_x 4 + 3 = 0 \Rightarrow (4\log_x 4 - 3)(6\log_x 4 - 1) = 0 \Rightarrow \log_x 4 = ..., ...$ They must obtain two values for their log.
A1	For both correct values. $\log_x 4 = \dfrac{3}{4}, \dfrac{1}{6}$
M1	For undoing either log correctly, but they **must** obtain a value for $x =$ ….. $x = 4^{\frac{4}{3}}$ or $x = 4^6$ Allow this mark for any erroneous log they find, but undo correctly.
A1	Accept $4^{\frac{4}{3}}$ or awrt 6.35 and 4^6 or 4096
Works in base 2	
M1	Changes the base of **at least one** log to base 2. $\dfrac{3}{2}\log_2 x$, $\dfrac{8\log_x x^6}{\log_2 x}$
M1	For applying the power law correctly seen anywhere in their work. This mark can also be awarded for explicit application of the power law on $\log_2 64 = 6\log_2 2$
M1	For multiplying through by $\log_2 x$ and forming a 3TQ in base 2 $\dfrac{3}{2}(\log_2 x)^2 - 22\log_2 x + 48 = 0 \Rightarrow \left[3(\log_2 x)^2 - 44\log_2 x + 96 = 0\right]$
M1	For solving their 3TQ by any valid and correct method. If there is no method seen with an incorrect 3TQ or with incorrect solutions following a correct 3TQ this is M0 $(12\log_2 x - 1)(8\log_2 x - 3) = 0 \Rightarrow \log_2 x = ..., ...$
A1	For both correct values. $\log_2 x = \dfrac{8}{3}, 12$
M1	For undoing either log correctly. $x = 2^{\frac{8}{3}}$ or $x = 2^{12}$ Allow this mark for any erroneous log they find, but undo correctly.
A1	Accept $2^{\frac{8}{3}}$ or awrt 6.35 and 2^{12} or 4096

Type MS · Log equations that turn into a quadratic

Answer 8 of 8 · Question 10 part whole · 7 marks

2023 JUN · Paper 02 (regional R) · extract 3 of 3 · find part whole in this mark scheme extract

Question	Scheme	Marks